OMER BESE

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SUMMARY

Energy systems engineer, founder, builder, operator. Started in the field on LADWP's commercial lighting rebate program, running projects from bulb counts to utility sign-off, then helped launch an industrial energy-audit program at the Istanbul Chamber of Industry. Founded Bonjuur in 2022 and raised $90K. Since 2026, building and shipping software with AI agents, mostly solo: WattShed, rebate software for residential energy efficiency (wattshed.co); Re_Tera, an insurance-policy app on the App Store; and ManorOS, a home energy auditor on the App Store. Built WattShed independently; now looking to build products like it inside a team.

EXPERIENCE

WATTSHED

July 2026 to Present

Founder and Engineer — Los Angeles

Community energy programs for data centers | wattshed.co

- Thesis: a rebate priced per kilowatt saved, set against the funder's own return rather than a utility's rate case; audited with vision AI rather than a truck roll; settled in weeks; and visible to the household on next month's bill. Target buyers: hyperscalers and data center developers, companies like Base Power, and government agencies and utilities.
- Built the first prototype solo at OpenAI Build Week (July 2026), then made it a product. A deterministic TypeScript engine applies the rules and AI structures the inspection evidence; a vitest suite asserts that every figure on the methodology page equals a live call to that engine.
- Built the program runtime, live in production: funders set the terms, households apply, contractors quote and document the work, reviewers clear it, and funders pay on verified results; the whole journey passed end to end on production in October 2026.
- Public, no-login program designer for all 50 states and DC (degree-day, modified-bin, and deemed-savings methods). A vision AI auditor estimates the house's current efficiency from video and pre-approves upgrades; a liveness check verifies that before-and-after video was captured live in that session.
- Split the platform into 40 modules across six suites; 30 run as public, no-login tools on wattshed.co. Built a live MCP server over the pricing engine (wattshed.co/mcp) and published the Mitchell County, Texas case study (modeled: 641 homes, about 2.06 MW of peak off the local grid).
- Stack: Next.js 16, React 19, TypeScript, Firebase, Google Cloud (Cloud Run, Cloud Tasks), OpenAI Responses API, MCP SDK; about 1,900 unit tests, Playwright end-to-end tests and GitHub Actions CI.

BONJUUR

April 2022 to December 2025

Founder — New York & Los Angeles

- Built a housekeeping service around short daily visits instead of the three-hour minimum. Cleaned houses personally every weekday for six months, more than 500 service visits, to experience the bottlenecks firsthand; built a 180-person waitlist.
- Raised $90,000, including $25,000 from Jason Calacanis after his pre-accelerator, getting him to break both of his rules: no solo founders, no service businesses. First customer: Jordan Fliegel, Managing Director of Techstars NYC.
- Pivot (2025): turned the service into AI estate management, with bodycams and vision AI to automate oversight of in-home service work, then refocused ManorOS on home energy.

TULIP HAUS

March 2020 to October 2025

Co-Founder — New York

- Co-founded a two-person business importing handmade, high-end Turkish home goods. Was set to sign a six-figure SoHo retail lease on March 12, 2020, but decided to wait on COVID; the city locked down eight days later. Pivoted to e-commerce, then to B2B wholesale as the exclusive US representative of a couple of Turkish brands.
- Ran the full chain with a co-founder: sourcing from artisans in Turkey, customs brokerage and tariff classification, ocean freight, inventory in NYC's Starrett-Lehigh Building, last-mile delivery and the Shopify store.

SPIROHOME

April 2019 to November 2019

Business Developer — Ankara, Turkey

- Oversaw FDA-clearance work for a battery-powered IoT medical device, coordinating hardware engineering, regulatory, and sales; the company obtained clearance during the role.
- Launched the company's Amazon UK and EU stores in the restricted "Medical Device II" category in three months, a process that usually takes around ten months, by preparing lithium-battery shipping (UN 38.3), CE marking, and EU medical-device compliance documentation before Amazon asked for it.

ISTANBUL CHAMBER OF INDUSTRY

January 2018 to March 2019

Associate Specialist, Energy & Environment — Istanbul, Turkey

- Helped develop and launch an industrial energy-audit program with Yıldız Technical University: the Chamber funded audit and measurement equipment, students audited member factories at no charge using the factories' real data, and professors oversaw the reports. Supervised the

program's first audit, at Sümer Plastik's 12,000 m² factory.

NULARIS

November 2016 to November 2017

Energy Efficiency Engineer Los Angeles

- Ran LADWP Commercial Lighting Incentive Program rebate projects end to end for a contractor on the utility's preferred list, from many small garages to the Jonathan Club, the Sofitel and the Four Seasons: fixture-by-fixture audits to California Title 24 and ASHRAE, savings in kW, kWh and GHG equivalents, LADWP pre-inspection, installation checks and post-installation audits before LADWP released funds.
- Personally replaced about half of the Jonathan Club's bulbs with LEDs when crews were short-staffed. Most clients saw at least a 30% reduction in lighting energy use.

PROJECTS

Built with AI agents, 2026. Solo except PowerUP (hackathon team of four).

PowerUP **Neighborhood islands for a home-battery fleet | October 2026 | github.com/mrbese/PowerUp**

InnovateLA Dual-Use Hackathon (USC), Energy and Grid Hardening track sponsored by Base Power, team of four; top 20 of the field and poster presentation on demo day. Owned the energy design and the build direction: an opt-in neighborhood island behind SCE's existing switch for the 456 homes behind switch RAR8848, a wildfire-shutoff dispatcher replayed on the real October 2026 Malibu shutoff timeline, and Interlock, a guard that passes every order to the fleet through eight checks, built from about 870 MB of SCE, CEC and CAISO data.

Re_Tera **Insurance-policy investigation and home-inventory app | 2026 to Present | getretera.com**

Reads a homeowner's or renter's policy, explains coverage in plain language, flags coverage gaps, sublimits and exclusions, and checks findings against a 50-state insurance-statute corpus with a citation validator. Native Swift/SwiftUI rewrite from first commit on June 5, 2026 to App Store release in three weeks (June 26, 2026).

ManorOS **Home energy auditor iOS app, live on the App Store | 2026 to Present | apps.apple.com/us/app/manoros/id6759580676**

LiDAR room scanning (ARKit, RoomPlan), simplified Manual J room load estimates, HVAC nameplate OCR, utility-bill and tariff checks, and upgrade recommendations with payback periods. Live on the App Store since October 2026.

Other projects **2026**

CellSense: home battery economics across four battery systems, 16 utilities and 35 rate plans, as a web calculator and a SwiftUI app. HazShip: lithium-battery shipping classification under 49 CFR 173.185, the IATA DGR and the IMDG Code, entirely on device. Shitster / DashOff: browser-only consumer document tools. BESE: a custom tokenizer for OpenAI Parameter Golf (PR #1327, 1.1276 bits per byte, self-reported). 1Relay: API key wallet with a zero-knowledge vault, and proxy for AI services.

EDUCATION

M.S. Sustainability Management | Columbia University **2015 to 2016**
New York

Capstone, Reducing Milk Waste in NYC with WWF: proposed UHT milk in place of refrigerated milk; WWF selected it for implementation.

B.S. Energy Systems Engineering | Istanbul Bilgi University **2010 to 2014**
Istanbul, Turkey

Thesis on geothermal power in the Seferihisar field, Izmir, built on proprietary well data obtained directly from Ormat Technologies.

SKILLS

ENERGY: building audits, utility rebate programs, deemed-savings, degree-day and modified-bin methods, simplified Manual J, demand-charge billing and tariffs, home-battery payback, battery dispatch, GIS on utility open data.

AI AND SOFTWARE: AI agents (Claude Code, Codex, Cursor) and multi-agent orchestration, vision AI, retrieval-augmented generation, MCP servers, TypeScript, Next.js, Swift/SwiftUI, Firebase, Google Cloud, Cloudflare, Playwright, Vitest, GitHub Actions.

OPERATIONS: field operations, importing and customs, e-commerce and B2B wholesale, regulated-category marketplace launches, FDA clearance.

WRITING

Geothermal Is in "Heat", and this time, buyer is moving in **mrbese.com | August 2026 | mrbese.com/writing/geothermal-heat**

Argues that geothermal was held back by transmission distance rather than geology, and that AI data centers can move to the wellhead.

Current War II: Tesla vs Edison **mrbese.com | July 2026 | mrbese.com/writing/war-of-the-currents**

Three-part series arguing that direct current is coming back, from native-DC loads to NEC Article 726 Class 4 fault-managed power.

Pila Energy: The Permissionless Insight, Not the Battery **mrbese.com | May 2026 | mrbese.com/writing/pila-energy**

Argues that Pila's advantage is regulatory, not the cell.

What I Learned About Base Power **mrbese.com | May 2026 | mrbese.com/writing/base-power**

Research essay on Base Power's ERCOT model from primary sources.